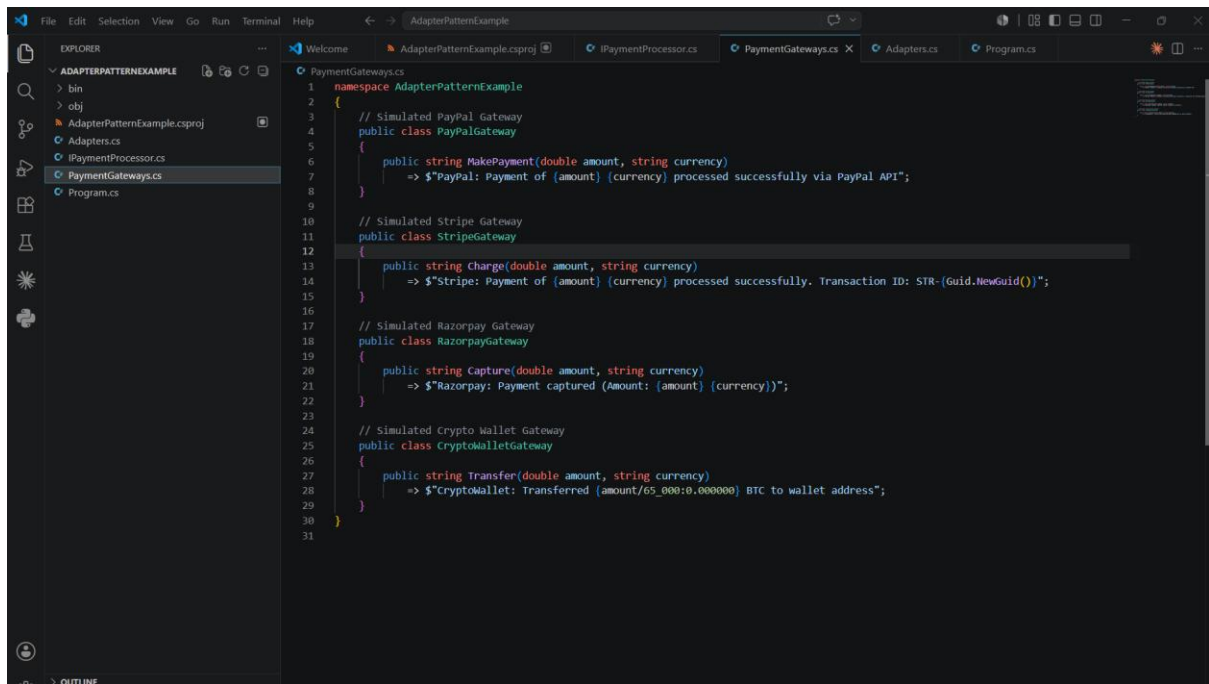


Hands-On Report

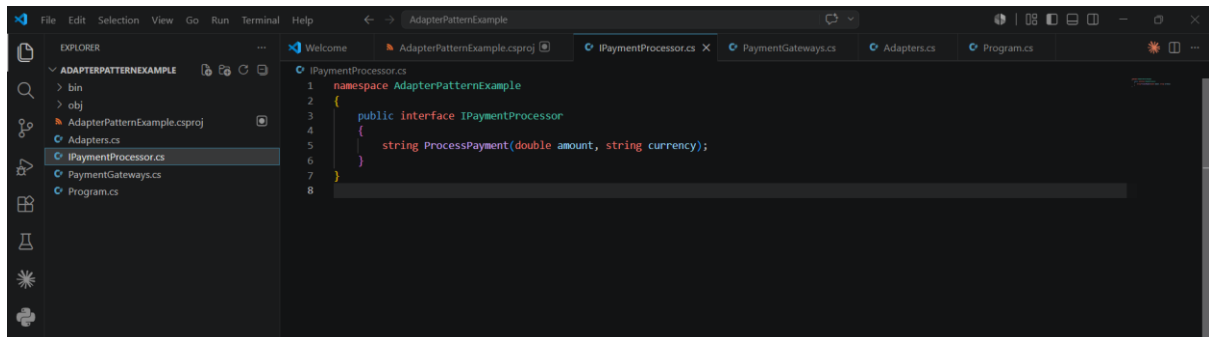
Exercise 4: Implementing the Adapter Pattern

In this exercise, I designed and tested the Adapter Pattern in a .NET application. I created a common **PaymentProcessor** interface and connected multiple payment gateway classes using separate adapter classes. Each adapter translated the standard payment request into the format required by its respective gateway, enabling all payment services to work through a single interface.

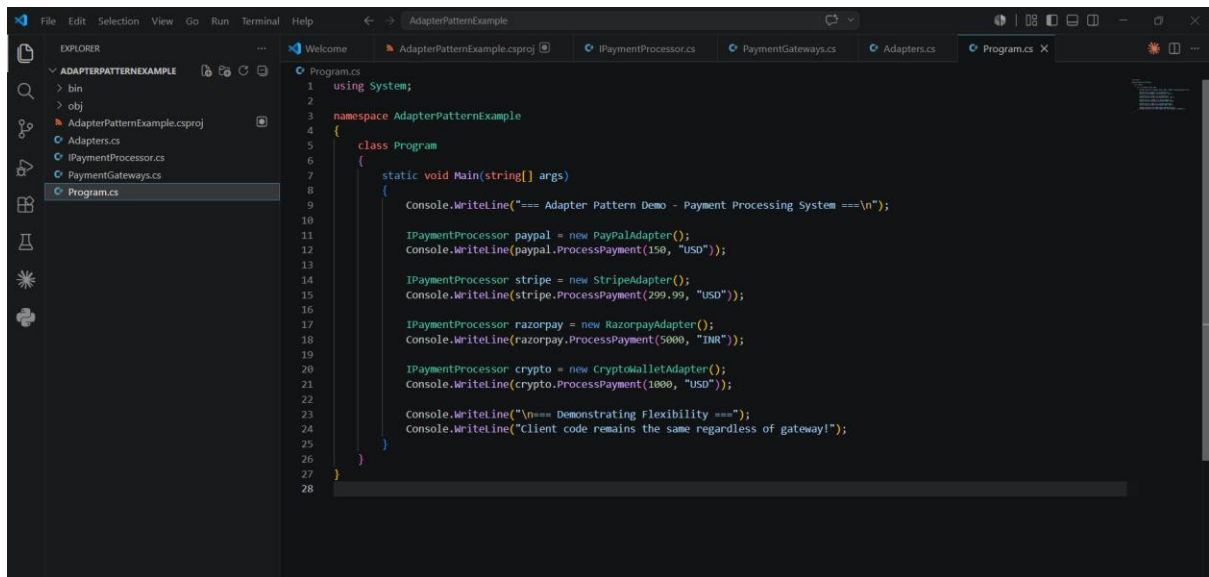
After completing the implementation, I executed the application and verified that different payment gateways processed transactions successfully using their adapters. This activity demonstrated how the Adapter Pattern bridges incompatible interfaces, promotes code reusability, and makes it easier to integrate external services while maintaining a consistent and flexible application design.



```
1 namespace AdapterPatternExample
2 {
3     // Simulated PayPal Gateway
4     public class PayPalGateway
5     {
6         public string MakePayment(double amount, string currency)
7         {
8             => $"PayPal: Payment of {amount} {currency} processed successfully via PayPal API";
9         }
10
11     // Simulated Stripe Gateway
12     public class StripeGateway
13     {
14         public string Charge(double amount, string currency)
15         {
16             => $"Stripe: Payment of {amount} {currency} processed successfully. Transaction ID: STR-{Guid.NewGuid()}";
17         }
18
19     // Simulated Razorpay Gateway
20     public class RazorpayGateway
21     {
22         public string Capture(double amount, string currency)
23         {
24             => $"Razorpay: Payment captured (Amount: {amount} {currency})";
25         }
26
27     // Simulated Crypto Wallet Gateway
28     public class CryptoWalletGateway
29     {
30         public string Transfer(double amount, string currency)
31         {
32             => $"CryptoWallet: Transferred {amount/65_000:0.000000} BTC to wallet address";
33         }
34     }
35 }
```

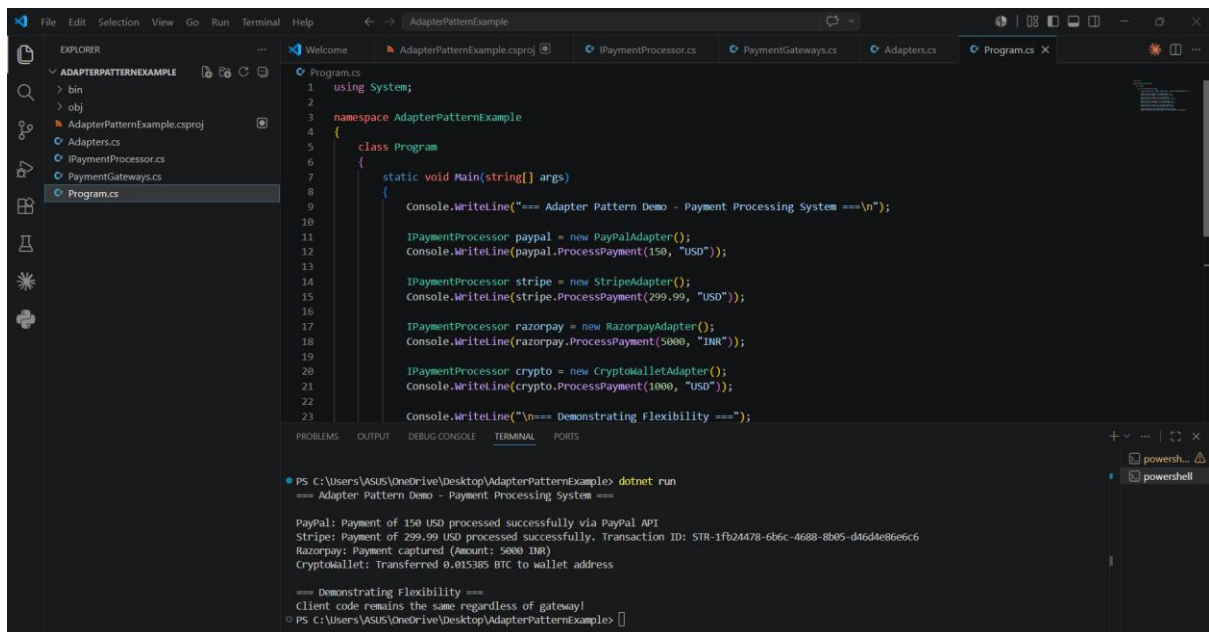


```
1 namespace AdapterPatternExample
2 {
3     public interface IPaymentProcessor
4     {
5         string ProcessPayment(double amount, string currency);
6     }
7 }
8
```



The screenshot shows the Visual Studio IDE with the AdapterPatternExample project open. The Explorer pane on the left shows the project structure. The main editor displays the Program.cs file with the following code:

```
1 using System;
2
3 namespace AdapterPatternExample
4 {
5     class Program
6     {
7         static void Main(string[] args)
8         {
9             Console.WriteLine("=== Adapter Pattern Demo - Payment Processing System ===\n");
10
11             IPaymentProcessor paypal = new PayPalAdapter();
12             Console.WriteLine(paypal.ProcessPayment(150, "USD"));
13
14             IPaymentProcessor stripe = new StripeAdapter();
15             Console.WriteLine(stripe.ProcessPayment(299.99, "USD"));
16
17             IPaymentProcessor razorpay = new RazorpayAdapter();
18             Console.WriteLine(razorpay.ProcessPayment(5000, "INR"));
19
20             IPaymentProcessor crypto = new CryptoWalletAdapter();
21             Console.WriteLine(crypto.ProcessPayment(1000, "USD"));
22
23             Console.WriteLine("\n=== Demonstrating Flexibility ===");
24             Console.WriteLine("Client code remains the same regardless of gateway!");
25         }
26     }
27 }
28
```



The screenshot shows the Visual Studio IDE with the AdapterPatternExample project open. The Explorer pane on the left shows the project structure. The main editor displays the Program.cs file. The bottom pane shows the output of the application, which is the result of running the program in a PowerShell terminal.

```
PS C:\Users\ASUS\OneDrive\Desktop\AdapterPatternExample> dotnet run
=== Adapter Pattern Demo - Payment Processing System ===

PayPal: Payment of 150 USD processed successfully via PayPal API
Stripe: Payment of 299.99 USD processed successfully. Transaction ID: STR-1fb24478-6b6c-4688-8b05-d46d4e8e6c6
Razorpay: Payment captured (Amount: 5000 INR)
CryptoWallet: Transferred 0.015385 BTC to wallet address

=== Demonstrating Flexibility ===
Client code remains the same regardless of gateway!
PS C:\Users\ASUS\OneDrive\Desktop\AdapterPatternExample>
```